

Unit 5: Big Data and Advanced Analytics

Complete Hinglish Revision Notes with Numerical Examples

1. Introduction to Big Data

Big Data ka matlab hai:

- extremely huge data
- fast generated data
- complex data

Traditional databases is data ko efficiently handle nahi kar paate.

Examples of Big Data

Platform	Data Generated
YouTube	Videos uploads
Instagram	Photos & reels
Amazon	Customer purchases
Google Maps	Live traffic
Banking	Transactions

2. Characteristics of Big Data (5 V's)

(1) Volume

Huge amount of data.

Example

Facebook:

- billions of posts
 - TBs/PBs data
-

(2) Velocity

Data generation ki speed.

Example

UPI transaction per second.

(3) Variety

Different forms of data.

Type	Example
Structured	Tables
Semi-Structured	JSON
Unstructured	Images/videos

(4) Veracity

Data accurate hai ya noisy?

Example

Fake product reviews.

(5) Value

Useful insights nikalna.

Example

Netflix recommendations.

3. Big Data Challenges

Challenge	Meaning
Storage	Huge data store karna
Processing	Fast computation
Security	Privacy protection

Challenge	Meaning
Scalability	Growing data handling
Integration	Different data combine

4. Real Life Applications

Netflix

Movie recommendation.

Amazon

Product suggestions.

Healthcare

Disease prediction.

Banking

Fraud detection.

Google Maps

Traffic prediction.

5. Hadoop

Apache Hadoop ek open-source framework hai jo:

- big data store karta hai
 - big data process karta hai
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Hadoop Features

Feature	Meaning
Distributed	Multiple systems
Scalable	Easy expansion

Feature	Meaning
Fault Tolerant	Failure recovery
Cost Effective	Cheap hardware

6. Hadoop Components

(1) HDFS

Storage system of Hadoop.

Large files ko blocks mein divide karta hai.

HDFS Components

NameNode

Metadata maintain karta hai.

DataNode

Actual data store karta hai.

(2) MapReduce

Processing framework of Hadoop.

Large tasks ko small parallel tasks mein divide karta hai.

Word Count Numerical Example

Input:

AI is good
AI is powerful

Map Phase

```
(AI,1)
(is,1)
(good,1)

(AI,1)
(is,1)
(powerful,1)
```

Shuffle & Sort

```
AI -> (1,1)
is -> (1,1)
good -> (1)
powerful -> (1)
```

Reduce Phase

```
(AI,2)
(is,2)
(good,1)
(powerful,1)
```

7. Hadoop Ecosystem

Tool	Purpose
Hive	SQL queries
Pig	Scripting
HBase	NoSQL database
Sqoop	Data transfer
Spark	Fast processing
Flume	Log collection
YARN	Resource manager

8. Apache Spark

Spark ek fast in-memory processing framework hai.

Spark Features

Feature	Meaning
In-memory processing	RAM based
Fast	Faster than MapReduce
Scalable	Easy expansion
Streaming	Real-time data

Spark Components

Component	Work
Spark Core	Main engine
Spark SQL	SQL processing
MLlib	Machine learning
Streaming	Live data

9. Apache Pig

Pig ek data flow scripting tool hai.

Language:

- Pig Latin
-

Example Pig Script

```
A = LOAD 'students.txt';
B = FILTER A BY marks > 50;
DUMP B;
```

10. Apache Hive

Hive SQL-like queries allow karta hai Hadoop data par.

Example Hive Query

```
SELECT *  
FROM students  
WHERE marks > 80;
```

11. Apache HBase

HBase ek distributed NoSQL database hai.

Used for:

- real-time read/write
-

HBase Features

Feature	Meaning
NoSQL	Non-relational
Distributed	Multiple machines
Scalable	Huge data support

12. Apache Sqoop

Sqoop relational database aur Hadoop ke beech data transfer karta hai.

Example

```
sqoop import  
--connect jdbc:mysql://localhost/db  
--table students
```

13. MapReduce Paradigm

MapReduce ek distributed processing model hai.

Main Steps

```
Input
  ↓
Map
  ↓
Shuffle & Sort
  ↓
Reduce
  ↓
Output
```

General Algorithm

```
MAP(key, value):
    process data
    emit(intermediate_key, value)

REDUCE(intermediate_key, values):
    combine values
    emit(result)
```

14. Matrix Vector Multiplication using MapReduce

Matrix

```
1 2
3 4
```


Vector

5
6

Step 1: Multiplication

Row 1

$$(1 \times 5) + (2 \times 6)$$

$$= 5 + 12$$

$$= 17$$

Row 2

$$(3 \times 5) + (4 \times 6)$$

$$= 15 + 24$$

$$= 39$$

Final Output Vector

17
39

MapReduce Approach

Mapper Output

(1, 5)
(1, 12)

(2,15)
(2,24)

Reducer Output

(1,17)
(2,39)

Important Comparison Table

Technology	Purpose
Hadoop	Big data framework
HDFS	Storage
MapReduce	Processing
Spark	Fast analytics
Hive	SQL querying
Pig	Data scripting
HBase	NoSQL database
Sqoop	Data transfer

Important Exam Definitions

Big Data

Large and complex datasets difficult to process using traditional tools.

Hadoop

Open-source framework for distributed storage and processing.

HDFS

Distributed file system of Hadoop.

MapReduce

Programming model for distributed processing.

Spark

Fast in-memory processing framework.

Hive

SQL-like data warehouse tool.

Pig

High-level scripting platform.

HBase

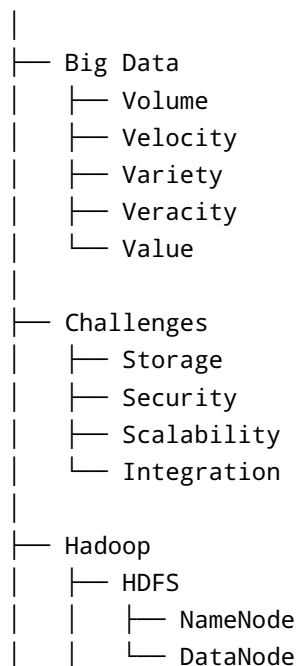
Distributed NoSQL database.

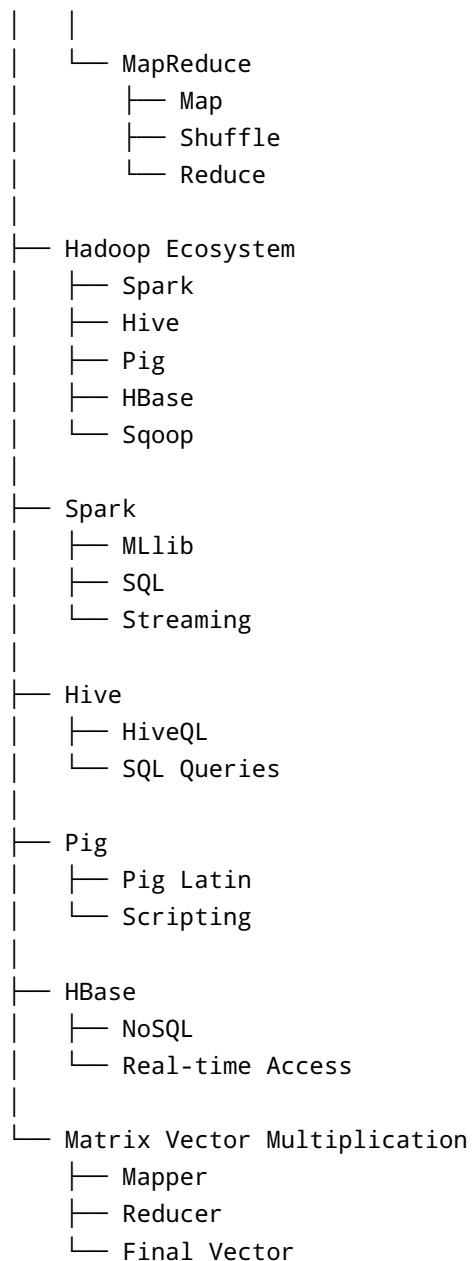
Sqoop

Tool for transferring data between databases and Hadoop.

Full Mind Map

BIG DATA & ADVANCED ANALYTICS





Super Quick Revision

Topic	Key Point
Big Data	Huge complex data
Hadoop	Distributed framework
HDFS	Storage
MapReduce	Processing
Spark	Fast analytics

Topic	Key Point
Hive	SQL queries
Pig	Data scripting
HBase	NoSQL
Sqoop	DB ↔ Hadoop transfer

Important Viva Questions

1. What are 5 V's of Big Data?
 2. Difference between Hadoop and Spark?
 3. Explain HDFS architecture.
 4. What is MapReduce?
 5. Difference between Pig and Hive?
 6. What is HBase?
 7. Explain Sqoop.
 8. Explain matrix vector multiplication using MapReduce.
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Short Tricks

5 V's

Volume Velocity Variety Veracity Value

Hadoop Ecosystem Memory Trick

"SPHHS"

S → Spark P → Pig H → Hive H → HBase S → Sqoop